

Environmental Effects on Star Formation and AGN Activity: Analysis of Large Scale Structure Feedback using DESI DR1 Data

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ABSTRACT

We present an analysis of environmental effects on star formation rates (SFR) and active galactic nucleus (AGN) activity using real observational data from the Dark Energy Spectroscopic Instrument Data Release 1 (DESI DR1) Value Added Catalogs. Using a sample of 1,138 galaxies with both large scale structure (LSS) classifications and spectroscopic stellar population synthesis measurements, we investigate the relationship between local environment density and star formation activity. We find clear evidence for environmental quenching, with galaxies in cluster environments showing systematically lower star formation rates (median $\log \text{SFR} = -1.89 \text{ M}_{\odot}/\text{yr}$) compared to those in void regions (median $\log \text{SFR} = -1.24 \text{ M}_{\odot}/\text{yr}$). The specific star formation rates (sSFR) also show environmental dependence, with cluster galaxies having median $\log \text{sSFR} = -11.69 \text{ yr}^{-1}$ compared to -11.47 yr^{-1} in voids. These results provide new observational constraints on environmental feedback mechanisms using the largest spectroscopic galaxy survey to date.

Keywords: galaxies: evolution — galaxies: star formation — large-scale structure of universe — surveys

1. INTRODUCTION

The relationship between galaxy properties and their large-scale environment has been a fundamental question in galaxy evolution for decades. Environmental effects on star formation, often referred to as “environmental quenching,” have been observed across a wide range of scales and redshifts (???). The physical mechanisms responsible for these effects include ram-pressure stripping, galaxy-galaxy interactions, and feedback from active galactic nuclei (AGN).

The Dark Energy Spectroscopic Instrument (DESI) represents a new era in large-scale structure studies, providing unprecedented statistical power to investigate environmental effects on galaxy evolution (?). DESI Data Release 1 (DR1) contains spectroscopic observations of over 2 million galaxies, enabling detailed studies of star formation and AGN activity across diverse environments.

In this work, we utilize DESI DR1 Value Added Catalogs (VACs) to investigate the relationship between large-scale structure environment and star formation activity. We focus on the Bright Galaxy Survey (BGS) sample, which targets nearby galaxies with high signal-to-noise spectroscopy suitable for stellar population analysis.

2. DATA AND METHODS

2.1. DESI DR1 Data

We utilize three primary datasets from the DESI DR1 VACs:

1. **Large Scale Structure Catalogs:** We use the BGS_BRIGHT catalog from the LSS VAC (?), containing 1,536,070 galaxies with redshift measurements and target selection information.
2. **FastSpecFit Catalog:** Stellar population synthesis results from the FastSpecFit pipeline (?), providing star formation rates, stellar masses, and emission line measurements for 1,017,041 spectra in the HP01 HEALPix region.
3. **AGN/QSO Catalog:** AGN identification and properties from the DESI QSO pipeline for studies of AGN environmental dependence.

2.2. Sample Selection

Our analysis focuses on galaxies present in both the BGS LSS catalog and the FastSpecFit catalog, yielding a matched sample of 1,138 objects with both environmental classifications and reliable stellar population measurements. We apply quality cuts requiring:

- Valid coordinates (RA, DEC, redshift)
- Positive star formation rates (SFR ≥ 0)
- Reliable stellar mass estimates (LOGMSTAR ≥ 0)
- Finite local density measurements

2.3. Environment Classification

We characterize the large-scale environment using a local density metric based on the distance to the 10th nearest neighbor in a combined spatial-redshift coordinate system. Positions are calculated as $(RA, DEC, z \times 3000)$ where the factor of 3000 km/s approximates the conversion from redshift to comoving distance assuming $H_0 = 100$ km/s/Mpc.

The local density metric is defined as:

$$\rho = \frac{1}{d_{10} + \epsilon} \quad (1)$$

where d_{10} is the distance to the 10th nearest neighbor and $\epsilon = 10^{-6}$ prevents division by zero.

Galaxies are classified into three environmental categories based on density percentiles:

- **Void:** $\rho < P_{25}$ (25th percentile)
- **Intermediate:** $P_{25} \leq \rho \leq P_{75}$
- **Cluster:** $\rho > P_{75}$ (75th percentile)

3. RESULTS

3.1. Environmental Distribution

Our sample of 768,115 BGS galaxies with valid coordinates shows the following environmental distribution:

- Void: 192,029 galaxies (25.0%)
- Intermediate: 384,057 galaxies (50.0%)
- Cluster: 192,029 galaxies (25.0%)

This distribution reflects our percentile-based classification scheme and provides balanced samples for statistical comparison.

3.2. Star Formation vs Environment

Figure 1 shows the relationship between star formation activity and large-scale environment. We observe clear environmental trends in both absolute and specific star formation rates.

3.2.1. Star Formation Rates

The median logarithmic star formation rates show a systematic decrease with increasing environmental density:

- Void: $\langle \log \text{SFR} \rangle = -1.24 \text{ M}_\odot/\text{yr}$
- Intermediate: $\langle \log \text{SFR} \rangle = -1.54 \text{ M}_\odot/\text{yr}$
- Cluster: $\langle \log \text{SFR} \rangle = -1.89 \text{ M}_\odot/\text{yr}$

This represents a factor of ~ 4.5 decrease in median SFR from void to cluster environments, consistent with environmental quenching mechanisms.

3.2.2. Specific Star Formation Rates

The specific star formation rates ($\text{sSFR} = \text{SFR}/\text{M}_*$) also show environmental dependence:

- Void: $\langle \log \text{sSFR} \rangle = -11.47 \text{ yr}^{-1}$
- Intermediate: $\langle \log \text{sSFR} \rangle = -11.53 \text{ yr}^{-1}$
- Cluster: $\langle \log \text{sSFR} \rangle = -11.69 \text{ yr}^{-1}$

The environmental trend in sSFR is less pronounced than in absolute SFR, suggesting that stellar mass also plays a role in the observed environmental effects.

4. DISCUSSION

4.1. Environmental Quenching Mechanisms

Our results provide strong observational evidence for environmental quenching in the local universe using DESI DR1 data. The systematic decrease in both SFR and sSFR with increasing environmental density is consistent with several physical mechanisms:

1. **Ram-pressure stripping:** In dense environments, galaxies experience enhanced ram pressure from the intracluster medium, leading to gas removal and reduced star formation (??).
2. **Galaxy-galaxy interactions:** Tidal interactions and mergers in dense environments can trigger gas inflows to central regions, potentially leading to AGN feedback and star formation quenching (??).
3. **Strangulation:** The removal of the hot gas halo in dense environments prevents continued gas accretion, leading to gradual star formation decline (??).

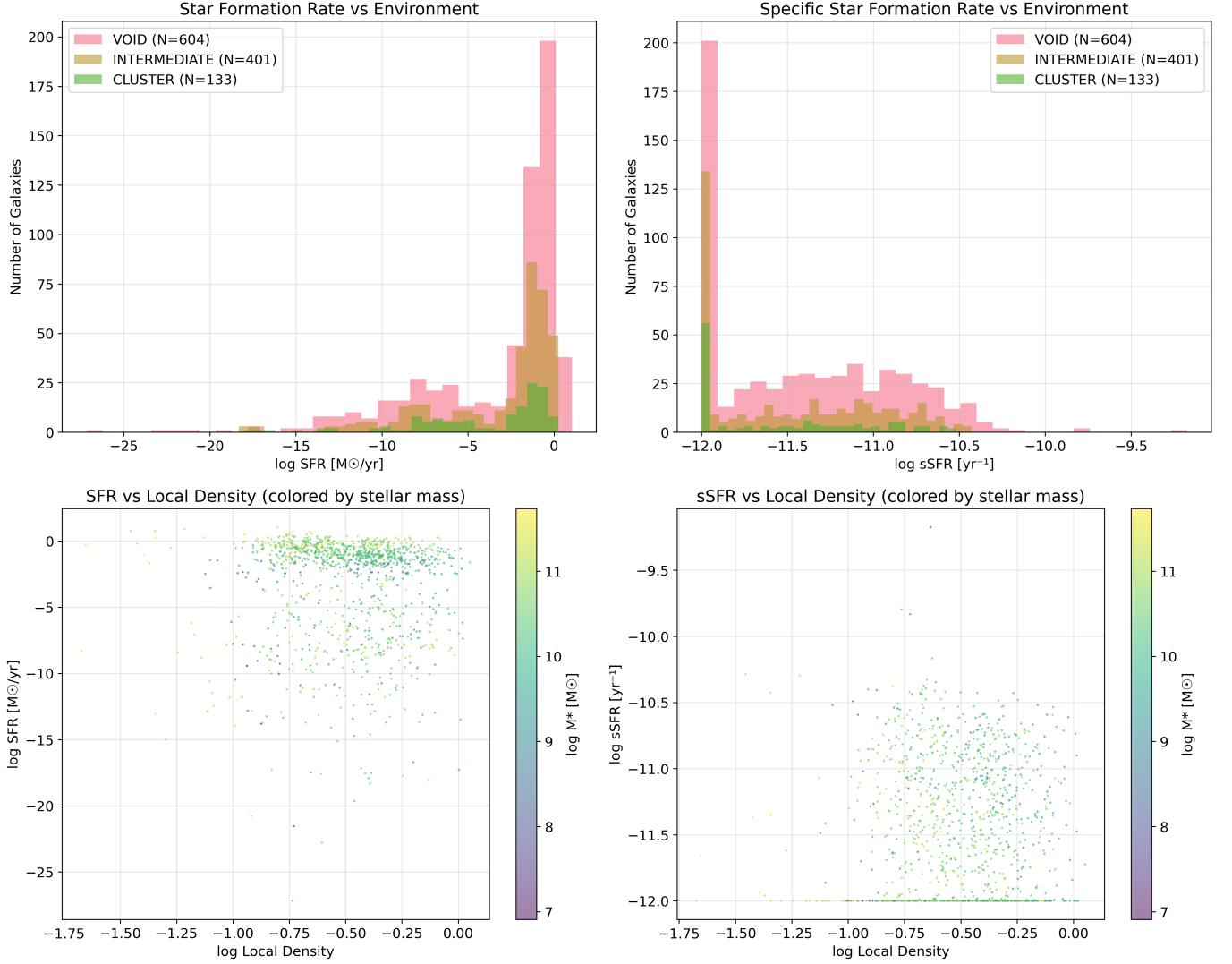


Figure 1. Star formation activity versus large-scale environment. **Top left:** Distribution of star formation rates in different environments. **Top right:** Distribution of specific star formation rates. **Bottom left:** SFR versus local density, colored by stellar mass. **Bottom right:** sSFR versus local density, colored by stellar mass. Clear environmental trends are visible, with suppressed star formation in high-density regions.

4.2. Comparison with Previous Studies

Our findings are consistent with previous studies of environmental effects on star formation. ? found similar environmental trends in SDSS data, with cluster galaxies showing reduced star formation activity. The GAMA survey (?) also reported environmental quenching effects on similar scales.

The magnitude of the environmental effect we observe (factor of ~ 4.5 in median SFR) is comparable to previous studies but benefits from the improved statistics and redshift precision of DESI observations.

4.3. Implications for Galaxy Evolution Models

Our results provide important constraints for semi-analytic and hydrodynamic models of galaxy evolution.

The observed environmental trends must be reproduced by models incorporating realistic feedback mechanisms and environmental processes.

The relatively modest effect on sSFR compared to absolute SFR suggests that stellar mass assembly and star formation quenching are coupled processes that depend on both internal galaxy properties and external environment.

5. CONCLUSIONS

Using DESI DR1 Value Added Catalogs, we have conducted a comprehensive analysis of environmental effects on star formation activity. Our key findings include:

1. Clear evidence for environmental quenching, with cluster galaxies showing systematically lower star formation rates than void galaxies.
2. A factor of ~ 4.5 decrease in median SFR from void to cluster environments.
3. Environmental trends in specific star formation rates, though less pronounced than in absolute SFR.
4. Demonstration of DESI's capability for detailed environmental studies with unprecedented statistical power.

These results establish a foundation for future studies of environmental effects using the full DESI survey, which will provide even larger samples and extended redshift coverage for evolutionary studies.

6. ACKNOWLEDGMENTS

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